

**HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF ELECTRICAL AND ELECTRONIC
ENGINEERING**



GRADUATION THESIS

**PEER-TO-PEER (P2P) ENERGY TRADING
FRAMEWORK FOR MULTI-MICROGRID SYSTEMS
USING ADMM-BASED DISTRIBUTED OPTIMIZATION
AND SECOND-ORDER CONE PROGRAMMING (SOCP)**

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Thesis title: *Peer-to-Peer (P2P) Energy Trading Framework for Multi-Microgrid Systems Using ADMM-Based Distributed Optimization and Second-Order Cone Programming (SOCP)*.

- I. Review the state-of-the-art in peer-to-peer (P2P) energy trading within multi-microgrid systems; analyze the role of distributed energy resources (DERs), physical grid constraints, and privacy requirements.
- II. Present the theoretical background on power flow models (from LinDistFlow to SOCP relaxation) and distributed optimization algorithms (ADMM).
- III. Develop a mathematical model for local microgrid energy management, including operational constraints for DGs, BESSs, renewable sources, and the objective function for cost minimization.
- IV. Propose a distributed P2P coordination algorithm combining ADMM consensus with dynamic penalty and SOCP relaxation; derive the locational marginal price (LMP) from ADMM dual variables.
- V. Simulate and analyze results for two case studies (3-MG and 4-MG cross-connected systems); evaluate economic performance, voltage regulation, network losses, and convergence behavior.

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ABSTRACT

The rapid penetration of renewable energy sources into distribution networks has driven the emergence of microgrids (MGs) as a fundamental building block of future smart grids. Multi-microgrid (Multi-MG) systems offer significant potential for peer-to-peer (P2P) energy trading, enabling economic dispatch and energy autonomy among prosumers. However, existing coordination frameworks commonly adopt a single-bus MG abstraction, which neglects the physical constraints of the distribution network—including voltage magnitude limits, line thermal limits, and the nonlinear coupling between active power, reactive power, and voltage (P-Q-V). Furthermore, centralized approaches compromise the privacy of individual microgrid operators.

This thesis proposes a fully distributed P2P energy trading framework for multi-microgrid systems that simultaneously addresses both the physical and informational challenges. The framework integrates: (1) a **Second-Order Cone Programming (SOCP) relaxation** of the Branch Flow Model (BFM) to accurately capture multi-bus AC network physics within each microgrid, and (2) an **ADMM (Alternating Direction Method of Multipliers) consensus algorithm with a dynamic penalty parameter** to decompose the global optimization problem into privacy-preserving local subproblems. A key feature of the proposed approach is that the **Locational Marginal Price (LMP)** emerges endogenously from the ADMM dual variables, providing a physically meaningful and cost-reflective pricing signal without requiring a centralized market operator.

Simulation results on two test cases—a three-microgrid radial system and a four-microgrid cross-connected network—demonstrate that the proposed framework achieves superior economic performance, maintains voltage profiles within permissible bounds, minimizes network losses, and converges reliably. Compared to single-bus and linearized (LinDistFlow) benchmarks, the SOCP-based approach eliminates physically infeasible dispatch schedules and provides accurate reactive power coordination.

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DECLARATION OF ORIGINALITY

I, **Trinh Xuan Thanh**, Student ID: [MSSV], Class: Electrical [XX] – K[XX], hereby declare that the work presented in this thesis, entitled “*Peer-to-Peer (P2P) Energy Trading Framework for Multi-Microgrid Systems Using ADMM-Based Distributed Optimization and Second-Order Cone Programming (SOCP)*”, is the product of my own research carried out under the supervision of Assoc. Prof. Dr. Nguyen Duc Tuyen. All data reported are truthful. All cited material complies with intellectual property regulations and is fully referenced. I accept full responsibility for the content of this thesis.

The Author

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LIST OF SYMBOLS AND ABBREVIATIONS

Abbreviation	Full Form
1. Methods & Algorithms	
ATC	Analytical Target Cascading
MPC	Model Predictive Control
OPF	Optimal Power Flow
AC-OPF	Alternating Current Optimal Power Flow
SOCP	Second-Order Cone Programming
ADMM	Alternating Direction Method of Multipliers
2. Energy Systems & Components	
MG	Microgrid
P2P	Peer-to-Peer
PV	Photovoltaic
WT	Wind Turbine
BESS	Battery Energy Storage System
DG	Diesel Generator
SOC	State of Charge
ESS	Energy Storage System
RES	Renewable Energy Sources
3. Metrics & Units	
VOLL	Value of Lost Load
pu	Per-unit
MWh	Megawatt-hour
MW	Megawatt

Note: Subscripts i, j denote node or line indices, and t denotes the time interval. They are omitted in descriptions for brevity.

Symbol	Description
1. Sets & Indices	
$\mathcal{T}$	Set of time scheduling intervals
$\mathcal{N}, \mathcal{M}$	Sets of buses/nodes in the network, and Microgrids
$\mathcal{C}, \mathcal{E}$	Sets of line configurations, and distribution lines
$\mathcal{N}^{PV}, \mathcal{N}^{WT}$	Sets of nodes equipped with PV and Wind Turbines
$\mathcal{N}^{DG}, \mathcal{N}^{BESS}$	Sets of nodes equipped with Diesel Generators and BESS
$\mathcal{N}^{trade}$	Set of nodes with P2P tie-lines
$\Omega(i)$	Set of child nodes connected to node i
2. Parameters	

S_{base}, U_{base}	System base apparent power and voltage
Z_{base}, I_{base}	System base impedance and current
R_{ij}, X_{ij}, C_{ij}	Resistance, reactance, and capacitance of line (i, j)
P_{load}, Q_{load}	Active and reactive power demand
π_{buy}, π_{sell}	Electricity prices for purchasing/selling to the main grid
$P_{PV}^{peak}, P_{WT}^{peak}$	Maximum active power generation of PV and WT
Q_{PV}, Q_{WT}	Reactive power output limits of PV and WT
S_{PV}, S_{WT}	Apparent power capacities of PV and WT
π_{trade}	P2P trading price between MGs
λ, ρ	Lagrange multiplier and penalty parameter for ADMM
P_{target}	Target active power trading amount provided by DSO
$\alpha_{DG}, \beta_{DG}, \gamma_{DG}$	Fixed, linear, and quadratic cost coefficients of DG
$C_{start, DG}$	Cold start cost of DG
$P_{DG}^{min}, P_{DG}^{max}$	Minimum and maximum active power limits of DG
R_{DG}^{ramp}	Ramp rate limit of DG
$T_{up}^{DG}, T_{down}^{DG}$	Minimum up and down time of DG
C_{BESS}, η_{bess}	Degradation cost coefficient and round-trip efficiency of BESS
P_{bess}^{max}	Maximum charging/discharging power of BESS
SOC^{min}, SOC^{max}	Minimum and maximum State of Charge limits of BESS
SOC^{ini}	Initial State of Charge of BESS
Cap_{bess}	Energy capacity of BESS

3. Decision Variables

P_{line}, Q_{line}	Active and reactive power flow on line (i, j)
V^2, I^2	Squared voltage magnitude and squared current magnitude
P_{PV}, P_{WT}	Scheduled active power generation of PV and WT
P_{DG}, Q_{DG}	Active and reactive power generation of DG
I_{DG}, v_{DG}	ON/OFF status and startup binary variables of DG
R_{DG}	Spinning reserve of DG
$P_{bess}^{char}, P_{bess}^{dis}$	Charging and discharging power of BESS
SOC	State of Charge of BESS
$I_{bess}^{char}, I_{bess}^{dis}$	Binary indicators for charging and discharging modes of BESS
P_{shed}	Total active power load shedding
$P_{shed}^{normal}, P_{shed}^{critical}$	Normal and critical load shedding amounts
P_{tie}	Active power traded via P2P tie-lines
$P_{grid}^{buy}, P_{grid}^{sell}$	Active power purchased from and sold to the main grid
$P_{grid}^{paid}, P_{grid}^{free}$	Compensated and uncompensated active power sold to the grid
$sSOC$	Slack variable for end-of-horizon SOC penalty

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CHAPTER 1. INTRODUCTION

1.1 Problem Statement: Power System Transition and Energy Coordination Challenges

In the context of global climate change and the depletion of fossil fuels, the traditional power system is undergoing a profound and comprehensive transition. Historically, power systems were designed around large, centralized fossil-fuel power plants that generated electricity and transmitted it over long distances to passive end-users. However, environmental concerns and technological advancements have catalyzed a paradigm shift. The massive penetration of Distributed Energy Resources (DERs)—such as solar photovoltaics (PV), wind turbines, and energy storage systems—has fundamentally altered the operational paradigm of conventional grids [1, 2]. Electricity consumers are no longer passive participants; they have transformed into prosumers capable of self-generating and injecting power back into the grid. This bidirectional flow of energy challenges the traditional unidirectional design of distribution networks, which were not originally built to handle dynamic, localized generation. Consequently, this decentralized evolution poses immense challenges in energy coordination. Integrating millions of DERs requires a flexible, decentralized management mechanism rather than traditional centralized dispatch, to prevent network congestion, manage bidirectional power flows, and avoid safety limit violations within the distribution grid [3].

1.2 From Centralized Grids to Microgrids: The Role of Distributed Energy Resources

To effectively manage the complexity introduced by DERs, the concept of the Microgrid (MG) has emerged as an essential structural solution [4]. Unlike the macrogrid, which operates on a massive scale and is vulnerable to cascading failures, microgrids allow for the localized clustering of loads and distributed generators within a defined electrical boundary. They are capable of operating either in grid-connected mode, drawing or supplying power to the main grid, or in islanded mode, functioning autonomously during grid outages [5]. These entities act as the fundamental building blocks of the Smart Grid, decentralizing energy management, significantly reducing transmission losses over long distances, and enhancing power supply reliability for local communities. Furthermore, the integration of Battery Energy Storage Systems (BESS) into microgrids effectively addresses the inherent intermittency and volatility of renewable energy sources by storing surplus generation during periods of high output and discharging it during peak load hours or times of low generation [6]. As the number of microgrids proliferates across the distribution network, the operational challenge expands from managing a single-microgrid scope to ensuring the optimal coordination and stable operation of intricate, interconnected multi-microgrid systems.

1.3 Peer-to-Peer (P2P) Energy Trading: Enhancing Economic Efficiency and Energy Autonomy

Instead of relying solely on one-way transactions with utility companies via fixed, often rigid pricing mechanisms like Feed-in-Tariffs (FIT), Peer-to-Peer (P2P)

energy trading introduces a disruptive and flexible economic model [7, 8]. P2P trading empowers prosumers and microgrids to directly trade their surplus energy with one another through localized Energy Markets [9]. This peer-driven approach democratizes the energy sector by bypassing traditional utility intermediaries, thereby reducing transaction overheads and empowering consumers. Recent literature demonstrates that, compared to conventional FIT mechanisms, P2P trading yields superior financial benefits for participants, significantly shortens the payback period for initial DER investments, and maximizes the overall social welfare of the participating community [10, 11]. This mechanism not only creates direct economic incentives for sustainable energy investments but also fosters the energy autonomy and resilience of local communities [12]. Nevertheless, realizing P2P trading in practice requires a highly robust mathematical and operational framework to effectively reconcile the economic objectives of diverse market participants with the stringent physical limitations and security constraints of the underlying grid infrastructure.

1.4 Challenges in Multi-Microgrid Networks

1.4.1 Limitations of Single-Bus Microgrid Abstraction

A critical flaw in the majority of existing literature on P2P energy markets is the widespread abstraction of an entire microgrid as a single electrical node (single-bus abstraction) or the assumption of a copper-plate network, where power can flow unhindered between any two points without losses or limits [13, 14]. While computationally convenient, this approach completely neglects the underlying physical constraints of the practical distribution network, such as actual line resistances, inherently radial topologies, and strict branch power capacity limits. When purely economically driven P2P transaction schedules are physically dispatched over the real network, they frequently lead to severe network congestion, critical voltage limit violations, and potential equipment damage [15, 16]. Even when researchers attempt to address this fundamental flaw by utilizing linearized power flow models (e.g., LinDistFlow or DC-OPF), the inherent approximation errors can reach 5-10% in systems with high renewable penetration, compromising the mathematical exactness required for safe and reliable grid operation [17, 18]. Consequently, there is an urgent need to employ rigorous Second-Order Cone Programming (SOCP) relaxations. These advanced techniques transform the highly non-linear and non-convex AC Optimal Power Flow (AC-OPF) problem into a mathematically tractable convex form, guaranteeing global optimality while rigorously preserving the grid's true physical reality and operational safety margins.

1.4.2 Importance of Joint Active-Reactive Power Co-optimization

Modern distribution networks typically exhibit high resistance-to-reactance (R/X) ratios, distinguishing them from transmission systems. Consequently, effective voltage regulation within these localized grids is heavily dependent not only on reactive power (Q) but also substantially on active power (P) flows. Despite this physical reality, many existing P2P trading studies focus exclusively on active power optimization to maximize direct financial returns, completely disregarding the vital reactive power injection and absorption capabilities of the advanced smart inverters coupled with DERs and BESS [19, 20]. Neglecting this crucial aspect severely diminishes the local voltage regulation capabilities, potentially leading to instability, and forces the system to forfeit valuable potential revenue from providing necessary ancillary services to

the main grid. The joint, holistic co-optimization of both active and reactive power for BESS and distributed generators has been conclusively proven to yield significantly higher overall economic efficiency while robustly safeguarding the distribution grid against localized voltage sags and power quality degradation [21–25].

1.4.3 Privacy Requirements and Distributed Pricing

Beyond the complex physical constraints of power flow, safeguarding data privacy represents arguably the most substantial non-technical barrier to the widespread scalability and adoption of interconnected multi-microgrid networks. Traditional Centralized Energy Management Systems (EMS) inherently require the continuous aggregation of highly sensitive operational information from all participating microgrids—including proprietary cost functions, internal network topologies, real-time load profiles, and specific BESS parameters—to a single, remote central server for optimization. This centralized architecture not only blatantly violates the operational privacy of independent microgrid operators but also creates a critical single point of failure, making the entire energy network highly vulnerable to malicious cyberattacks and data breaches [26, 27]. To effectively overcome this critical vulnerability, the Alternating Direction Method of Multipliers (ADMM) algorithm has been widely adopted due to its exceptional ability to preserve absolute data privacy. ADMM achieves optimal coordination by requiring only the iterative exchange of limited boundary power variables and price signals at the point of common coupling, without ever revealing internal parameters [28, 29]. Furthermore, to accurately and fairly reflect the true operational cost of energy at each specific location—explicitly accounting for both physical network losses and potential line congestion—the Distribution Locational Marginal Pricing (DLMP) mechanism, which can be extracted endogenously directly from the ADMM dual variables, proves vastly superior to traditional, overly simplistic uniform pricing schemes [30–32].

1.5 Research Motivation and Objectives

A critical and systematic review of the current literature reveals a clear and persistent research gap: very few studies have successfully integrated a comprehensive, holistic framework that simultaneously encompasses (1) an exact and rigorous physical grid model via SOCP relaxation to ensure operational safety, (2) a fully distributed, privacy-preserving ADMM-based solver featuring dynamic penalty parameters for accelerated algorithmic convergence, and (3) an endogenous, transparent DLMP pricing mechanism specifically tailored for multi-microgrid systems with deeply integrated BESS [33–37].

The primary motivation of this thesis is to definitively bridge this identified gap by developing a robust, unified model for modern decentralized energy networks. The specific research objectives formulated to achieve this overarching goal are as follows:

- To develop a comprehensive, mathematically rigorous energy management model for individual prosumers and microgrids utilizing the Branch Flow Model and SOCP relaxation techniques.
- To design a fully distributed, secure P2P trading algorithm based on the ADMM framework equipped with a dynamic penalty mechanism, thereby guaranteeing 100% operational data privacy and improving computational efficiency.

- To propose and implement an endogenous DLMP mechanism seamlessly derived from the ADMM convergence process to transparently compensate participants for actual network losses and actively alleviate localized grid congestion.
- To extensively simulate, test, and validate the proposed algorithmic performance, critically evaluating both the tangible techno-economic impacts and the convergence stability on complex, interconnected multi-microgrid test systems under various operational scenarios.

CHAPTER 2. THEORETICAL BACKGROUND AND EVOLUTION OF MATHEMATICAL MODELS

2.1 Evolution of Power Flow Models

2.1.1 *Classical Models: Single-Node and Linearized DistFlow*

The development of energy management models for microgrids has historically relied on simplified power flow representations. The most fundamental approach is the single-node abstraction, which aggregates all generations and loads into a single bus [13, 14]. While computationally trivial and suitable for high-level economic dispatch, this model fails to capture internal grid physics. By completely ignoring the network topology, it masks potential line overloads and internal voltage violations, leading to schedules that may be physically unrealizable when subjected to actual network constraints [15, 16]. To introduce spatial awareness without sacrificing linear solvability, the Linearized Distribution Flow (LinDistFlow) model was widely adopted [17, 18]. This model simplifies the power flow equations by ignoring transverse voltage drops and assuming negligible line losses, thereby providing a purely linear approximation of power flows [19, 20]. This linearity was highly attractive as it allowed the use of robust Mixed-Integer Linear Programming solvers. However, its accuracy severely deteriorates in heavily loaded networks or those with substantial reactive power flows. The lossless assumption becomes invalid when bidirectional power flows from high renewable energy penetration increase system losses, rendering it inadequate for modern active distribution networks where precise voltage regulation is mandatory [21, 22, 38, 39].

2.1.2 *Modern Model: Branch Flow Model and SOCP Relaxation*

To overcome the inaccuracies of classical approximations, the exact AC Optimal Power Flow (AC-OPF) modeled via the nonlinear Branch Flow Model (BFM) is required. This exact formulation precisely accounts for line losses, phase angles, and complex power flows. Nevertheless, the non-convex nature of AC-OPF, characterized by quadratic terms and trigonometric relationships, makes it computationally intractable and prone to trapping solvers in suboptimal local optima. Consequently, the Second-Order Cone Programming (SOCP) relaxation has emerged as the state-of-the-art methodology to bridge the gap between physical accuracy and computational feasibility [40]. By relaxing the rigorous nonlinear equality constraint linking active power, reactive power, voltage, and current into a mathematically convex inequality cone, SOCP convexifies the entire problem [19, 41]. This vital transformation guarantees that any local optimum found is also the global optimum, ensuring both global optimality and computational efficiency while rigorously respecting the underlying physics of radial distribution networks [42–44]. Under standard operating conditions—such as avoiding significant reverse power flows and maintaining sensible voltage limits—the SOCP relaxation is mathematically proven to be exact, meaning the relaxed inequality constraint naturally becomes strictly active at the optimal solution, yielding physically meaningful and exact AC results.

2.2 Significance of Reactive Power and Voltage in Distribution Networks

2.2.1 *Nonlinear P-Q-V Coupling in High R/X Ratio Networks*

Unlike transmission grids where the decoupling of active power-angle ($P-\delta$) and reactive power-voltage ($Q-V$) is a valid assumption due to high line reactance, distribution networks possess fundamentally different physical properties, primarily characterized by high resistance-to-reactance (R/X) ratios. In such resistive environments, active power flows cause significant voltage drops, inducing a strong, nonlinear coupling between active power (P), reactive power (Q), and voltage magnitudes (V) [23, 24]. This means that any fluctuation in active power generation from Distributed Energy Resources (DERs) or dynamic power exchanges in peer-to-peer (P2P) markets will simultaneously and directly impact local voltage profiles [45]. Consequently, the traditional paradigm of separating real power scheduling from voltage control is no longer viable. Therefore, energy management systems cannot treat active power trading merely as an isolated economic variable; it must be continuously and rigorously evaluated against its reactive power implications to proactively manage network congestion and maintain strict voltage stability across the entire distribution network [46].

2.2.2 *Trade-off Between Sequential Coordination and Co-optimization*

Traditional energy management frameworks often employ sequential coordination: active power dispatch is solved first to prioritize purely economic minimization, followed by a secondary reactive power adjustment step designed to correct any resulting voltage deviations. This decoupled approach is fundamentally suboptimal because the initial active power dispatch may exhaust the capacity of inverters or push the network state too close to its limits, leaving insufficient margins for the subsequent reactive power adjustments. As a result, this often forces the curtailment of renewable generation or leads to unnecessarily high operational costs to restore grid stability [25, 47]. In contrast, the joint co-optimization of P and Q explicitly integrates both power components into a single objective function and solves their operational constraints simultaneously. By deeply leveraging the often-underutilized reactive power capabilities of Battery Energy Storage Systems (BESS) and smart PV inverters, this holistic co-optimization approach maximizes the overall economic utility of physical assets. It effectively mitigates active power network losses and pre-emptively prevents voltage violations before they occur, proving fundamentally indispensable for the reliable and robust operation of modern multi-microgrid systems.

2.3 Evolution of Grid Optimization Algorithms

2.3.1 *Centralized Optimization and Privacy Barriers*

Historically, multi-microgrid energy management was orchestrated by a centralized Energy Management System (EMS), which actively gathers all detailed network parameters, real-time load demands, internal generator cost functions, and local physical constraints into a single computational hub [26, 27]. While centralized solvers can theoretically and readily identify the true global optimum due to complete system visibility, they exhibit critical real-world vulnerabilities. The strict prerequisite of sharing highly sensitive operational data and proprietary commercial information fundamentally violates the data sovereignty and privacy expectations of independent microgrid

operators, often deterring participation in cooperative frameworks [28, 48]. Furthermore, a fully centralized architecture inherently suffers from severe scalability bottlenecks; as the number of active DERs, smart meters, and dynamic prosumers scales exponentially, the required communication bandwidth and the central computational burden rapidly become unsustainable. This heavy reliance on a central server also exacerbates system vulnerability, introducing a critical single-point-of-failure that could compromise the entire multi-microgrid network during cyber-attacks or communication outages [29, 49].

2.3.2 Decentralized Optimization: ADMM and the Significance of Dual Variables

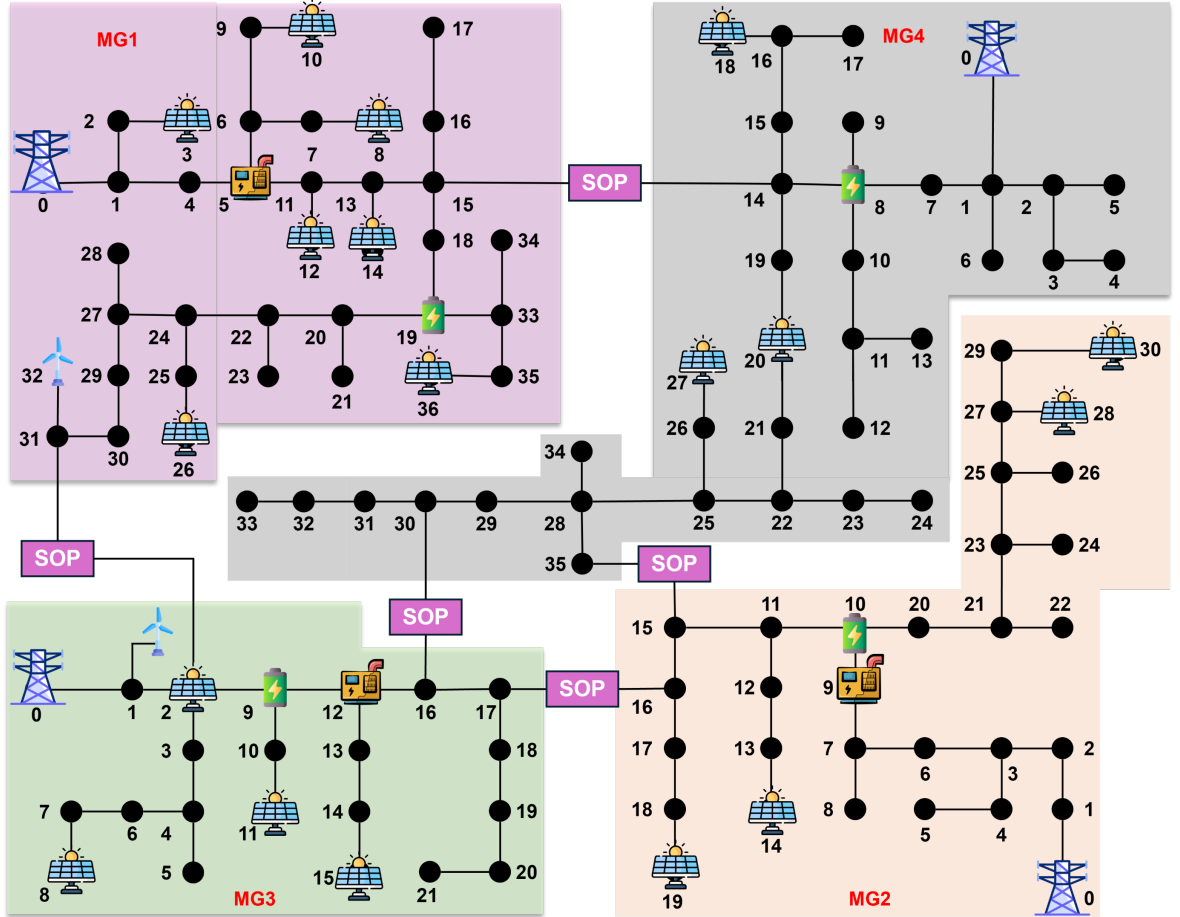
To resolve the inherent paradox between achieving global mathematical optimality and preserving local data privacy, decentralized optimization algorithms have become a paramount necessity in modern grid operations. Among various distributed techniques, the Alternating Direction Method of Multipliers (ADMM) stands out due to its superior numerical convergence properties, robustness against communication delays, and overall architectural elegance compared to standard dual ascent methods [50–52]. ADMM systematically decomposes the monolithic global coordination problem into smaller, mathematically independent subproblems that can be solved locally in parallel by each individual microgrid EMS. During this iterative process, the only information exchanged between neighboring entities is the expected power flow at their shared boundary nodes (the point of common coupling) alongside a corresponding dual variable, thereby ensuring the absolute privacy of all internal operational parameters and cost functions [53, 54]. Crucially, the dual variables utilized in ADMM are not merely abstract mathematical artifacts necessary for algorithmic convergence; they possess a profound physical and economic meaning. At the point of convergence, these dual variables intrinsically represent the Distribution Locational Marginal Price (DLMP) at the specific boundary nodes [30, 31]. This endogenous DLMP organically acts as a decentralized, dynamic pricing signal that transparently reflects the true, real-time cost of energy provision. By encapsulating not only the marginal cost of generation but also the marginal costs associated with ohmic network losses and line congestion, DLMPs automatically penalize grid-stressing transactions and naturally incentivize energy trading that supports overall system stability [32, 55–57].

CHAPTER 3. SYSTEM ARCHITECTURE AND MATHEMATICAL FORMULATION

3.1 Peer-to-Peer Interconnection Topology

The physical architecture of the proposed energy management system comprises one Main Grid (MG0) and four interconnected Microgrids (MG1, MG2, MG3, and MG4). These Microgrids (MGs) form a Peer-to-Peer (P2P) trading network, establishing a cooperative environment that facilitates active power exchange. This P2P interconnection is specifically designed to enhance system resilience, allowing the MGs to support one another through internal power transfers, especially when isolated from the Main Grid. To ensure grid stability and respect physical tie-line capacities, the power exchange between any connected MGs is strictly constrained within a trading limit of ± 1.5 MW (1.5 pu, given a system base power of 1 MVA).

The overarching network topology is illustrated in Figure 3.1. The distinctive physical characteristics and load-generation profiles of each MG are systematically detailed as follows:



Hình 3.1 Peer-to-Peer interconnection topology of the multi-microgrid system.

- **Microgrid 1 (MG1) - Heavy Load with Diverse Generation:** Comprising 36 nodes, MG1 operates as a heavily loaded center within the network. It features a diverse generation portfolio, integrating multiple Renewable Energy Sources (RES) including Photovoltaic (PV) systems and a Wind Turbine (WT), alongside

a Battery Energy Storage System (BESS) and a Diesel Generator (DG). MG1 maintains physical P2P tie-lines with MG3 and MG4.

- **Microgrid 2 (MG2) - Solar-Only Provider:** Characterized by a 30-node radial network, MG2 is a solar-only MG, devoid of wind generation. Thanks to its concentrated PV capacity, it acts as the major energy provider to the entire P2P network during the daytime. It incorporates BESS and DG units for stability and is directly connected to MG3 and MG4.
- **Microgrid 3 (MG3) - Light Load Savior:** As the smallest network with 21 nodes, MG3 operates with relatively light load demands but possesses excess generation capabilities. This structural advantage allows MG3 to act as the primary savior for the network, frequently exporting surplus power to support neighboring MGs. It is interconnected with MG1, MG2, and MG4.
- **Microgrid 4 (MG4) - Dense Complex with High PV:** With a total of 35 nodes, MG4 is defined by both heavy load demands and high PV capacity. It contains 25 active load nodes, making it a dense industrial and residential complex with massive power consumption. However, it heavily relies on four highly distributed PV nodes. This specific configuration causes aggressive power fluctuations, as the massive consumption drastically outweighs generation during non-sunlight hours. MG4 is interconnected with MG1, MG2, and MG3 to mitigate these aggressive internal imbalances.

3.2 System Parameters, Sets, and Indices

To ensure realistic simulations and validate the proposed mathematical models, the system parameters and input datasets are constructed based on physical configurations. The foundational global parameters and component data specifications are detailed as follows:

- **Global System Settings:** The multi-microgrid network is normalized using a system base apparent power (S_{base}) of 1 MVA. The operational scheduling is performed over a day-ahead time horizon denoted by $\mathcal{T}$, comprising 24 intervals with a discrete time step of $\Delta t = 1$ hour.
- **Peer-to-Peer Trading Limits:** To respect the physical thermal capacities of the interconnected cables, the active power exchanged via tie-lines ($P_{tie,i,t}$) between interconnected microgrids is strictly bounded within a trading limit of ± 1.5 MW (equivalent to 1.5 pu).
- **Value of Lost Load (VOLL):** In severe emergency scenarios, load shedding may be unavoidable. However, to guarantee the absolute protection of critical loads, the penalty cost (VOLL) associated with critical load shedding ($P_{shed,i,t}^{critical}$) is assigned a strict penalty (e.g., 100,000 cents/MWh). This heavy penalty strictly forces the optimization engine to ensure 100% continuous supply to critical demands.
- **Input Data Profiles and Network Attributes:** The dynamic 24-hour profiles for active ($P_{load,i,t}$) and reactive ($Q_{load,i,t}$) power demand, alongside the maximal

generation capacities for renewable sources ($P_{PV,i,t}^{\text{peak}}$ and $P_{WT,i,t}^{\text{peak}}$), are derived from physical load and source datasets. Furthermore, the structural network parameters, including the resistance (R_{ij}) and reactance (X_{ij}) of all internal distribution lines and tie-lines, are directly derived from detailed line configuration data to accurately reflect the real-world network topology.

To formulate the mathematical models of the proposed multi-microgrid system, several key sets and indices are defined. The operational time horizon is discretized into a set of intervals denoted by $\mathcal{T}$. Let $\mathcal{M}$ denote the set of all microgrids in the network. Within the system, the distribution network is represented by a set of nodes $\mathcal{N}$ and a set of connecting distribution lines (branches) $\mathcal{E}$. The specific sets of nodes equipped with distributed energy resources are defined as follows: $\mathcal{N}^{PV}$ for Photovoltaic systems, $\mathcal{N}^{WT}$ for Wind Turbines, $\mathcal{N}^{DG}$ for Diesel Generators, and $\mathcal{N}^{BESS}$ for Battery Energy Storage Systems.

Furthermore, the set of boundary nodes that possess tie-lines enabling physical Peer-to-Peer active power trading with neighboring microgrids is denoted by $\mathcal{N}^{\text{trade}}$. For the radial topology of the distribution network, $\Omega(i)$ represents the set of all child nodes directly connected to a parent node i , and $\mathcal{C}$ defines the set of available line configurations. Throughout the formulation, subscripts i and j refer to spatial node or branch indices, whereas the subscript t refers to the discrete time step.

3.3 Mathematical Modeling of Energy Components

This section details the mathematical formulation governing the operational constraints of the various energy components deployed within the microgrid network.

3.3.1 Renewable Energy Sources and Loads

The integration of RES, specifically PV systems and WT, alongside power loads, constitutes the primary non-dispatchable elements of the network. The active power output from PV and WT units is strictly bounded by their available forecasted generation peaks at any given time step t . The active power limits are formulated as follows:

$$0 \leq P_{PV,i,t} \leq P_{PV,i,t}^{\text{peak}}, \quad \forall i \in \mathcal{N}^{PV}, \forall t \quad (3.1a)$$

$$0 \leq P_{WT,i,t} \leq P_{WT,i,t}^{\text{peak}}, \quad \forall i \in \mathcal{N}^{WT}, \forall t \quad (3.1b)$$

where $P_{PV,i,t}$ and $P_{WT,i,t}$ denote the scheduled active power of the PV and WT at node i during time t , respectively. $P_{PV,i,t}^{\text{peak}}$ and $P_{WT,i,t}^{\text{peak}}$ represent their maximum available active power based on weather conditions.

For the load demand, both active ($P_{L,i,t}$) and reactive ($Q_{L,i,t}$) power are treated as fixed parameters unless load shedding is activated during emergency operations.

3.3.2 Battery Energy Storage System (BESS)

The BESS provides essential flexibility by charging during surplus periods and discharging during generation shortages. To prevent simultaneous charging and discharging, binary indicators $u_{ch,i,t}$ and $u_{dis,i,t}$ are introduced. The operational constraints for the BESS are defined as:

$$0 \leq P_{ch,i,t} \leq u_{ch,i,t} P_{BESS}^{\max}, \quad \forall i \in \mathcal{N}^{BESS}, \forall t \quad (3.2a)$$

$$0 \leq P_{dis,i,t} \leq u_{dis,i,t} P_{BESS}^{\max}, \quad \forall i \in \mathcal{N}^{BESS}, \forall t \quad (3.2b)$$

$$u_{ch,i,t} + u_{dis,i,t} \leq 1, \quad \forall i \in \mathcal{N}^{BESS}, \forall t \quad (3.2c)$$

where $P_{ch,i,t}$ and $P_{dis,i,t}$ are the charging and discharging power, respectively, bounded by the maximum capacity $P_{BESS}^{\max}$.

The state-of-charge (SOC) of the battery updates dynamically across consecutive time intervals, constrained by the energy capacity (E_{BESS}) and the round-trip efficiency (η_{BESS}):

$$SOC_{i,t} = SOC_{i,t-1} + \frac{P_{ch,i,t} \eta_{BESS} - P_{dis,i,t} / \eta_{BESS}}{E_{BESS}}, \quad \forall t > 0 \quad (3.3a)$$

$$SOC^{\min} \leq SOC_{i,t} \leq SOC^{\max}, \quad \forall t \quad (3.3b)$$

$$SOC_{i,0} = SOC_{i,24} = SOC^{\text{ini}} \quad (3.3c)$$

Equation (3.3c) ensures the energy neutrality of the BESS over a 24-hour daily cycle, setting the initial and final SOC to a predefined level SOC^{ini} .

3.3.3 Dispatchable Generator (DG)

DGs operate as controllable power sources, governed by their minimum and maximum output boundaries. The operational status of the DG is indicated by a binary variable $u_{DG,i,t}$. To ensure the technical limits of the Diesel generators, additional constraints including reactive power limits, startup status, minimum operating times, and spinning reserve are defined as follows:

$$u_{DG,i,t} P_{DG}^{\min} \leq P_{DG,i,t} \leq u_{DG,i,t} P_{DG}^{\max}, \quad \forall i \in \mathcal{N}^{DG}, \forall t \quad (3.4a)$$

$$(P_{DG,i,t})^2 + (Q_{DG,i,t})^2 \leq (S_{DG}^{\max})^2 u_{DG,i,t}, \quad \forall i \in \mathcal{N}^{DG}, \forall t \quad (3.4b)$$

$$u_{DG,i,t} Q_{DG}^{\min} \leq Q_{DG,i,t} \leq u_{DG,i,t} Q_{DG}^{\max}, \quad \forall i \in \mathcal{N}^{DG}, \forall t \quad (3.4c)$$

$$v_{DG,i,t} \geq u_{DG,i,t} - u_{DG,i,t-1}, \quad \forall i \in \mathcal{N}^{DG}, \forall t \quad (3.4d)$$

$$\sum_{\tau=t}^{t+T_{up}^{DG}-1} u_{DG,i,\tau} \geq T_{up}^{DG} v_{DG,i,t}, \quad \forall i \in \mathcal{N}^{DG}, \forall t \quad (3.4e)$$

$$\sum_{\tau=t}^{t+T_{down}^{DG}-1} (1 - u_{DG,i,\tau}) \geq T_{down}^{DG} (u_{DG,i,t-1} - u_{DG,i,t}), \quad \forall i \in \mathcal{N}^{DG}, \forall t \quad (3.4f)$$

$$R_{DG,t} = \sum_{i \in \mathcal{N}^{DG}} (u_{DG,i,t} P_{DG}^{\max} - P_{DG,i,t}), \quad \forall t \quad (3.4g)$$

$$R_{DG,t} \geq 0.2 \sum_{i \in \mathcal{N}^{DG}} P_{DG}^{\max}, \quad \forall t \quad (3.4h)$$

where $P_{DG,i,t}$ and $Q_{DG,i,t}$ are the active and reactive power of the DG, respectively, and $S_{DG}^{\max}$ represents its maximum apparent power. $v_{DG,i,t}$ denotes the startup state, and $R_{DG,t}$ represents the spinning reserve. The minimum up and down times are defined by T_{up}^{DG} and T_{down}^{DG} .

Additionally, the variation in the DG's power output between consecutive time intervals is bounded by its ramp-up and ramp-down limits to prevent mechanical stress:

$$-R_{DG} \leq P_{DG,i,t} - P_{DG,i,t-1} \leq R_{DG}, \quad \forall t > 0 \quad (3.5)$$

where R_{DG} is the maximum allowable ramping rate.

3.4 AC Optimal Power Flow Formulation

The core mathematical engine for the multi-microgrid system relies on the non-linear Alternating Current Optimal Power Flow (AC-OPF) model. Operating at the foundational level of the hierarchy, the local AC-OPF problem is formulated to robustly handle both standard operation and unexpected fault scenarios (such as generator failures, extreme renewable drops, or main grid blackouts). To manage this, the system utilizes a binary emergency indicator $\Gamma_E(t) \in \{0, 1\}$ to seamlessly toggle between normal operation ($\Gamma_E(t) = 0$) and an emergency fault state ($\Gamma_E(t) = 1$), ensuring operational continuity and optimal economic dispatch under any physical condition.

3.4.1 Economic and Loss Functions

The primary objective of the local microgrid operator is to minimize the total operational cost over the scheduling horizon $\mathcal{T}$. To generalize the optimization problem across varying grid conditions, the fundamental objective function unifies standard economic costs and load shedding penalty costs.

The total operational cost function $\mathcal{F}^{OPF}$ is expressed as:

$$\mathcal{F}^{OPF} = F_{grid} + F_{DG} + F_{BESS} + F_{loss} + F_{shed} \quad (3.6)$$

The individual economic and operational cost components are detailed as follows:

$$F_{grid} = \sum_{t \in \mathcal{T}} \left(\pi_{buy,t} P_{grid,t}^{buy} + \pi_{sell,t} P_{grid,t}^{sell} \right) \quad (3.7a)$$

$$F_{DG} = \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{N}^{DG}} \left(\alpha_{DG} u_{DG,i,t} + (\beta_{DG} + \kappa_{DG}) P_{DG,i,t} + \gamma_{DG} (P_{DG,i,t})^2 + C_{start,DG} v_{DG,i,t} \right) \quad (3.7b)$$

$$F_{BESS} = \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{N}^{BESS}} C_{BESS} \left(\eta_{BESS} P_{ch,i,t} + \frac{P_{dis,i,t}}{\eta_{BESS}} \right) \quad (3.7c)$$

$$F_{loss} = \sum_{t \in \mathcal{T}} \sum_{(i,j) \in \mathcal{E}} \omega_{loss} I_{ij,t}^2 \quad (3.7d)$$

$$F_{shed} = \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{N}} \left(C_{shed}^{normal} P_{shed,i,t}^{normal} + C_{shed}^{critical} P_{shed,i,t}^{critical} \right) \quad (3.7e)$$

Here, F_{grid} represents the cost of purchasing electricity and the revenue from selling power to the main grid. F_{DG} represents the operational cost of Diesel Generators, encompassing fixed, linear, and quadratic fuel costs, linear Operation and Maintenance (O&M) costs (represented by coefficient κ_{DG}), and startup costs. F_{BESS} calculates the degradation cost of the battery storage system proportional to its charging and discharging energy. To promote localized consumption and minimize distribution losses, F_{loss} acts as a virtual penalty on the squared current magnitude flowing through the distribution lines, scaled by a factor ω_{loss} .

Finally, F_{shed} leverages static, prohibitively high VOLL multipliers (C_{shed}^{normal} , $C_{shed}^{critical}$) to strictly penalize load curtailment and prioritize critical load preservation.

3.4.2 Operational Modes and Fault Scenarios

Rather than embedding fault logic directly into the physical equations (such as the DistFlow boundaries), the system isolates unexpected events as explicit operational constraints. Depending on the operational state, denoted by a binary emergency indicator $\Gamma_E(t) \in \{0, 1\}$, specific mathematical conditions are enforced to reflect either standard operation or component failures.

To protect against severe power deficits during faults, the load shedding limits are strictly defined by their physical capacity. Additionally, to maintain a constant power factor, the reactive power shed is constrained to be strictly proportional to the active power shed:

$$P_{shed,j,t} = P_{shed,j,t}^{normal} + P_{shed,j,t}^{critical} \quad (3.8a)$$

$$0 \leq P_{shed,j,t}^{normal} \leq 0.8P_{load,j,t} \quad (3.8b)$$

$$0 \leq P_{shed,j,t}^{critical} \leq 0.2P_{load,j,t} \quad (3.8c)$$

$$Q_{shed,j,t} = P_{shed,j,t} \frac{Q_{load,j,t}}{P_{load,j,t}} \quad (3.8d)$$

The transition between modes is mathematically handled as follows:

- **Normal Mode** ($\Gamma_E(t) = 0$): The microgrid operates with all components fully available. Due to the prohibitive VOLL penalties ($C_{shed}^{critical} \gg C_{shed}^{normal} \gg \text{cost}_{buy}$) in the objective function (3.7e), the shedding variables are naturally forced to zero by the optimizer. The system maintains its standard security margins, including the required spinning reserve:

$$R_{DG,t} \geq 0.2 \sum_{i \in \mathcal{N}^{DG}} P_{DG}^{\max} \quad (3.9)$$

- **Emergency Mode** ($\Gamma_E(t) = 1$): The system experiences a severe disruption, which could encompass a main grid blackout, a local generator trip, or an extreme drop in renewable generation. The failed component k is mathematically isolated by forcing its power output and exchange variables to zero:

$$P_{k,t} = 0, \quad Q_{k,t} = 0 \quad \forall k \in \Omega_{fault} \quad (3.10)$$

where Ω_{fault} represents the set of components experiencing an outage (e.g., the main grid interconnection, specific DGs, or PV units).

To counteract the resulting power deficit, the standard spinning reserve constraint (3.9) is intentionally relaxed. This relaxation allows the remaining dispatchable generators to fully utilize their 20% locked reserve capacity to maintain grid stability. If the internal generation remains insufficient, the solver is permitted to shed load according to (3.8b)-(3.8c), strongly guided by the economic prioritization in F_{shed} to preserve critical loads.

3.4.3 Distribution Power Flow Constraints

The power flow within the radial network of the microgrids is modeled using the original non-linear Branch Flow Model (DistFlow) equations. The active and reactive power balance, voltage drop, and branch current definitions are formulated as follows:

$$P_{line,ij,t} - R_{ij}I_{ij,t}^2 + P_{DG,j,t} + P_{PV,j,t} + P_{WT,j,t} + P_{dis,j,t} - P_{ch,j,t} + P_{tie,j,t} + P_{shed,j,t} - P_{load,j,t} = \sum_{k \in \Omega(j)} P_{line,jk,t} \quad \forall j \in \mathcal{N}, \forall t \in \mathcal{T} \quad (3.11a)$$

$$Q_{line,ij,t} - X_{ij}I_{ij,t}^2 + Q_{DG,j,t} + Q_{shed,j,t} - Q_{load,j,t} = \sum_{k \in \Omega(j)} Q_{line,jk,t} \quad \forall j \in \mathcal{N}, \forall t \in \mathcal{T} \quad (3.11b)$$

$$V_{j,t}^2 = V_{i,t}^2 - 2(R_{ij}P_{line,ij,t} + X_{ij}Q_{line,ij,t}) + (R_{ij}^2 + X_{ij}^2)I_{ij,t}^2 \quad \forall (i,j) \in \mathcal{E}, \forall t \in \mathcal{T} \quad (3.11c)$$

$$I_{ij,t}^2 = \frac{P_{line,ij,t}^2 + Q_{line,ij,t}^2}{V_{i,t}^2} \quad \forall (i,j) \in \mathcal{E}, \forall t \in \mathcal{T} \quad (3.11d)$$

$$(V^{\min})^2 \leq V_{j,t}^2 \leq (V^{\max})^2 \quad \forall j \in \mathcal{N}, \forall t \in \mathcal{T} \quad (3.11e)$$

$$0 \leq I_{ij,t}^2 \leq (I^{\max})^2 \quad \forall (i,j) \in \mathcal{E}, \forall t \in \mathcal{T} \quad (3.11f)$$

To correctly model the interaction with the main grid, a boundary condition is enforced at the slack bus (denoted as node $j = 0$). The branch power injected into the root node from the external network explicitly represents the grid power exchange:

$$P_{line,0,t} = P_{grid,t}^{buy} + P_{grid,t}^{sell} \quad \forall t \in \mathcal{T} \quad (3.12a)$$

$$Q_{line,0,t} = Q_{grid,t} \quad \forall t \in \mathcal{T} \quad (3.12b)$$

Equations (3.11a) and (3.11b) delineate the active and reactive power balance at node j at time t , respectively. The sum of all active power injected into node j —including the incoming branch power ($P_{line,ij,t}$), distributed generation ($P_{DG,j,t}$), renewable active power generation ($P_{PV,j,t}, P_{WT,j,t}$), BESS net discharge ($P_{dis,j,t} - P_{ch,j,t}$), P2P trading ($P_{tie,j,t}$), and load shedding ($P_{shed,j,t}$)—must equal the sum of the power flowing out to child branches $\sum P_{line,jk,t}$, the local load demand ($P_{load,j,t}$), and the line active power losses ($R_{ij}I_{ij,t}^2$). By convention, active power sold to the grid ($P_{grid,t}^{sell}$) takes non-positive values.

Similarly, the reactive power balance in (3.11b) accounts for branch flows ($Q_{line,ij,t}$), DG reactive support ($Q_{DG,j,t}$), and reactive load shedding ($Q_{shed,j,t}$). Notably, the renewable energy inverters (PV and WT) are configured to operate at unity power factor, contributing zero reactive power to the system.

Equation (3.11c) governs the voltage drop along branch (i, j) , considering the line impedance (R_{ij}, X_{ij}) and branch flows. Equation (3.11d) represents the non-linear definition of the squared branch current. Finally, inequalities (3.11e) and (3.11f) ensure that the squared nodal voltages and branch currents remain within their secure operational limits. Due to the non-linear and non-convex nature of the current definition in (3.11d), this equation is typically relaxed using Second-Order Cone Programming (SOCP) in optimization solvers.

3.5 SOCP Relaxation and Exactness

3.5.1 Convexification and SOC Relaxation

The non-linear nature of the AC Optimal Power Flow fundamentally stems from the current magnitude definition in the Branch Flow Model. As originally stated in Equation (3.11d), the squared branch current is formulated as:

$$I_{ij,t}^2 = \frac{P_{line,ij,t}^2 + Q_{line,ij,t}^2}{V_{i,t}^2}$$

This equality constraint is highly non-convex. To achieve a computationally tractable convex model, this equation is relaxed into a standard Second-Order Cone (SOC) inequality constraint. The SOC relaxation transforms the non-convex feasible region into a convex cone by replacing the equality with an inequality:

$$I_{ij,t}^2 V_{i,t}^2 \geq P_{line,ij,t}^2 + Q_{line,ij,t}^2 \quad (3.13)$$

3.5.2 Exactness in Normal Mode

The fundamental premise of substituting an equality with an inequality is predicated on the concept of exactness—the condition where the optimizer natively forces the inequality (3.13) to hold with equality. In the **Normal Mode** of operation, this exactness is inherently guaranteed. As defined in the objective function (3.6), the formulation incorporates a dedicated loss penalty component, F_{loss} (3.7d). Because the primary objective of the solver is to minimize the total operational cost, it actively suppresses the squared current variable $I_{ij,t}^2$, strictly driving it towards its theoretical lower bound. Consequently, the inequality holds with equality ($I_{ij,t}^2 V_{i,t}^2 = P_{line,ij,t}^2 + Q_{line,ij,t}^2$), thereby guaranteeing the exactness of the SOCP model and accurately reflecting the true physical state of the distribution grid.

3.5.3 Robustness against Inexactness in Emergency Mode

While exactness is guaranteed under standard operations, the mathematical literature suggests that significant challenges can arise during extreme **Emergency Modes**. A typical vulnerability occurs during islanded operations with a significant surplus of renewable energy generation. In standard models lacking curtailment mechanisms, if

the microgrid cannot export this excess power, the mathematical solver might erroneously exploit the SOCP relaxation gap. It could artificially inflate the current variable $I_{ij,t}^2$ to burn the surplus energy as fictitious line losses, violating the exactness ($I_{ij,t}^2 V_{i,t}^2 > P_{line,ij,t}^2 + Q_{line,ij,t}^2$) and rendering the results physically meaningless.

However, the proposed microgrid formulation is intrinsically designed to effectively counteract this phenomenon. The renewable generation constraints natively allow for free active power curtailment (3.1a). Confronted with excess generation, the solver will preferentially curtail the renewable output at zero cost rather than inflating $I_{ij,t}^2$, which would incur a strict financial penalty via the F_{loss} term scaled by ω_{loss} . This dual mechanism—free curtailment capability combined with a strict virtual loss penalty—ensures that the solver is mathematically prevented from exploiting the relaxation gap. As a result, the model robustly preserves the physical integrity and exactness of the network’s power flow equations even amidst extreme emergency and over-generation scenarios.

3.6 Final Local SOCP Optimization Model

Integrating the operational cost components from Section 3.4.1 and the relaxed physical network constraints, the final local optimization problem is formulated as a Second-Order Cone Program (SOCP). Although the problem is implemented and solved as a single unified code using the emergency indicator $\Gamma_E(t)$, from a mathematical presentation perspective, the active objective function and constraints can be categorized into two distinct formulations based on the operational state.

3.6.1 Formulation I: Normal Operation Mode ($\Gamma_E(t) = 0$)

In standard grid-connected mode, load shedding is economically penalized to the point of avoiding load curtailment ($F_{shed} \rightarrow 0$). The objective is purely focused on economic costs and loss minimization. The problem is defined as:

$$\begin{aligned}
\min \quad & \mathcal{F}_{normal}^{OPF} = F_{grid} + F_{DG} + F_{BESS} + F_{loss} \\
\text{s.t.} \quad & \text{Component Constraints (3.1a) – (3.5)} \\
& \text{Standard Spinning Reserve (3.9)} \\
& \text{DistFlow Boundary Conditions (3.12)} \\
& \text{Power Balance \& Voltage Constraints (3.11a) – (3.11c), (3.11e) – (3.11f)} \\
& \text{SOCP Relaxed Current Constraint (3.13)}
\end{aligned} \tag{3.14}$$

3.6.2 Formulation II: Emergency Mode ($\Gamma_E(t) = 1$)

When a generalized fault occurs, power generation or exchange with the affected components $k \in \Omega_{fault}$ completely ceases. Consequently, the operational costs associated with these specific failed components are eliminated from the objective function (as their power outputs are forced to zero). The primary goal of the remaining microgrid transitions from pure economic profit to system resilience, leveraging the unlocked reserve capacity and prioritizing critical load preservation. The problem

transitions to:

$$\begin{aligned}
\min \quad & \mathcal{F}_{emergency}^{OPF} = F_{grid} + F_{DG} + F_{BESS} + F_{loss} + F_{shed} \\
\text{s.t.} \quad & \text{Component Constraints (3.1a) – (3.5)} \\
& \text{Load Shedding Limits (3.8a) – (3.8d)} \\
& \text{Operational Fault Constraints (3.10)} \\
& \text{DistFlow Boundary Conditions (3.12)} \\
& \text{Power Balance \& Voltage Constraints (3.11a) – (3.11c), (3.11e) – (3.11f)} \\
& \text{SOCP Relaxed Current Constraint (3.13)}
\end{aligned} \tag{3.15}$$

CHAPTER 4. SPATIAL COORDINATION - ANALYTICAL TARGET CASCADING

4.1 Analytical Target Cascading (ATC) Overview

The operation of multi-microgrid (Multi-MG) systems entails a complex spatial coordination problem. Traditional centralized optimization methodologies require individual Microgrids (MGs) to transmit all their local, potentially sensitive information—such as load demand, renewable generation profiles, and Battery Energy Storage System (BESS) constraints—to a central operator. This paradigm not only creates computational bottlenecks for large-scale systems but also severely compromises the privacy of independent MG operators.

To resolve these physical and informational challenges, this framework adopts Analytical Target Cascading (ATC). ATC is a distributed optimization technique originally developed for multidisciplinary design optimization, which perfectly aligns with the hierarchical structure of modern distribution networks. By decomposing the global optimal power flow problem, ATC divides the Multi-MG system into a two-level hierarchical architecture:

- **Coordinator Level (Upper Level):** The Distribution System Operator (DSO) or Utility Grid serves as the central coordinator. The DSO assigns the optimal target variables $P_{target,ij,t}$ (conceptually acting as the ideal tie-line reference) and price signals to the local MGs to ensure Peer-to-Peer (P2P) consensus.
- **Local Microgrid Level (Lower Level):** Each individual MG operates as an independent autonomous entity. Utilizing the target signals provided by the coordinator, each MG solves its local Second-Order Cone Programming (SOCP) problem to determine its optimal internal asset dispatch and the scheduled tie-line power exchange $P_{tie,ij,t}$.

Through an iterative exchange of limited boundary variables (the target power $P_{target,ij,t}$ and the tie-line response $P_{tie,ij,t}$) alongside penalty multipliers, the ATC algorithm guides the multi-microgrid system toward a globally optimal equilibrium without exposing any internal constraints or local asset states. For mathematical clarity regarding directional power flow, the active power traded via the tie-line connecting MG i to MG j at time t is denoted as $P_{tie,ij,t}$.

4.2 The Coordinator Problem (Utility Grid)

At the top level of the ATC hierarchy, the Coordinator (Utility Grid) focuses entirely on optimizing the Peer-to-Peer (P2P) consensus. The coordinator's optimization problem formulates the target variable $P_{target,ij,t}$ that the lower-level MGs must track, yielding the optimal target $P_{target,ij,t} = (P_{tie,ij,t} - P_{tie,ji,t}) / 2$.

The objective function of the coordinator is formulated to penalize the mismatch between the coordinator's target $P_{target,ij,t}$ and the previous iteration's optimal tie-line scheduled power $P_{tie,ij,t}^{*,k}$ submitted by the local MGs, subject to strict reciprocity

($P_{target,ij,t} + P_{target,ji,t} = 0$). The optimization problem is given by:

$$\begin{aligned} \min \quad & \sum_{t \in \mathcal{T}} \sum_{(i,j) \in \mathcal{L}^{trade}} \left[\lambda_{ij,t}^k \left(P_{tie,ij,t}^{*,k} - P_{target,ij,t} \right) + \frac{\rho_{ij,t}^k}{2} \left(P_{target,ij,t} - P_{tie,ij,t}^{*,k} \right)^2 \right] \\ \text{s.t.} \quad & P_{target,ij,t} + P_{target,ji,t} = 0 \end{aligned} \quad (4.1)$$

where:

- $P_{target,ij,t}$ is the decision variable representing the optimal trading target determined by the coordinator for the power flow from node i to node j at time t .
- $P_{tie,ij,t}^{*,k}$ is the fixed local tie-line power exchange from i to j scheduled by the lower-level MG at the current iteration k .
- $\lambda_{ij,t}^k$ is the linear Lagrange multiplier (shadow price) for active power trading at iteration k .
- $\rho_{ij,t}^k$ is the quadratic penalty multiplier at iteration k , strictly penalizing the spatial deviation between the target and the local response.

After solving the optimal target $P_{target,ij,t}^{*,k}$ and receiving the updated $P_{tie,ij,t}^{*,k}$ from the lower-level MGs, the coordinator updates the penalty multipliers for the next iteration to drive the deviation to zero. The update equations for the multipliers (λ and ρ) are defined as follows:

$$\lambda_{ij,t}^{k+1} = \lambda_{ij,t}^k + \rho_{ij,t}^k \left(P_{tie,ij,t}^{*,k} - P_{target,ij,t} \right) \quad (4.2a)$$

$$\rho_{ij,t}^{k+1} = \begin{cases} \tau \rho_{ij,t}^k & \text{if } (r_{ij,t}^k)^2 > \mu (s_{ij,t}^k)^2 \\ \rho_{ij,t}^k / \tau & \text{if } (s_{ij,t}^k)^2 > \mu (r_{ij,t}^k)^2 \\ \rho_{ij,t}^k & \text{otherwise} \end{cases} \quad (4.2b)$$

where the primal residual (mismatch) is defined as $r_{ij,t}^k = P_{tie,ij,t}^{*,k} + P_{tie,ji,t}^{*,k}$, and the dual residual (speed) is defined as $s_{ij,t}^k = \rho_{ij,t}^k \left| P_{tie,ij,t}^{*,k} - P_{tie,ij,t}^{*,k-1} \right|$. The parameters are set as $\tau = 1.5$ and $\mu = 10$. This Adaptive Residual Balancing approach dynamically adjusts the penalty parameter to maintain a balance between primal and dual feasibility, significantly accelerating the convergence of the Analytical Target Cascading algorithm.

4.3 The Local Microgrid Problem with ATC

At the lower level of the hierarchical Analytical Target Cascading (ATC) architecture, the local Microgrids (MGs) receive the updated penalty multipliers ($\lambda_{ij,t}^k, \rho_{ij,t}^k$) and the optimal reference target ($P_{target,ij,t}^{k-1}$) broadcasted by the central Coordinator (Utility Grid). To align their local schedules with the global consensus, each MG integrates an Augmented Lagrangian penalty function into its base optimization objective.

The base objectives of each MG, defined as $\mathcal{F}_{normal}^{OPF}$ for grid-connected modes and $\mathcal{F}_{emergency}^{OPF}$ for islanded states (as previously established in Section 3.6), are mathematically penalized for any spatial discrepancy. The Augmented Lagrangian penalty

function corresponding to the active power deviation along the tie-lines is formulated as follows:

$$F_{penalty,i}^{ATC} = \sum_{t \in \mathcal{T}} \sum_{j \in \mathcal{N}_i} \left[\lambda_{ij,t}^k \left(P_{tie,ij,t} - P_{target,ij,t}^{k-1} \right) + \frac{\rho_{ij,t}^k}{2} \left(P_{tie,ij,t} - P_{target,ij,t}^{k-1} \right)^2 \right] \quad (4.3)$$

where $\mathcal{N}_i$ represents the set of neighboring Microgrids directly connected to MG i .

The incorporation of this penalty effectively bridges the gap between the decentralized operation and the global targets. The newly modified local ATC objective function, which each MG i aims to minimize autonomously, is therefore defined as:

$$\mathcal{F}_{ATC,i}^{OPF} = \mathcal{F}_i^{OPF} + F_{penalty,i}^{ATC} \quad (4.4)$$

where $\mathcal{F}_i^{OPF}$ represents either $\mathcal{F}_{normal}^{OPF}$ or $\mathcal{F}_{emergency}^{OPF}$ depending on the prevailing operational condition of the network for MG i .

Consequently, each autonomous MG solves this penalized Second-Order Cone Programming (SOCP) model subject to its complete set of local operational and physical constraints defined in Chapter 3. Upon convergence of its local optimization, the MG transmits only the resulting scheduled tie-line active power exchange, $P_{tie,ij,t}$, back to the Utility Grid Coordinator for the subsequent ATC iteration.

4.4 ATC Iterative Algorithm and Convergence

The execution of the Analytical Target Cascading (ATC) algorithm is an iterative process that alternates between solving the local MG optimization problems and updating the coordinator's target variables and penalty multipliers. The step-by-step procedure at iteration k is detailed as follows:

Step 1: Initialization. Set the iteration counter $k = 1$. Initialize the Lagrange multipliers $\lambda_{ij,t}^1 = 0$, the penalty parameters $\rho_{ij,t}^1 = \rho_0$, the initial tie-line power schedules $P_{tie,ij,t}^0 = 0$, and the initial targets $P_{target,ij,t}^0 = 0$ for all directional tie-lines $(i, j) \in \mathcal{L}^{trade}$ and time intervals $t \in \mathcal{T}$.

Step 2: Local Microgrid Optimization. Each MG autonomously solves its penalized optimal power flow problem as defined in (4.4), utilizing the target variables from the previous iteration $P_{target,ij,t}^{k-1}$. The optimal tie-line active power exchange $P_{tie,ij,t}^k$ is determined and subsequently transmitted to the Coordinator.

Step 3: Coordinator Update. Upon receiving the tie-line power schedules from all interacting MGs, the Coordinator updates the target variables, multipliers, and adaptive penalty parameters to enforce P2P consensus. First, the primal residual $r_{ij,t}^k$ and dual residual $s_{ij,t}^k$ are calculated to measure the spatial mismatch and the algorithm's progress:

$$r_{ij,t}^k = P_{tie,ij,t}^k + P_{tie,ji,t}^k \quad (4.5a)$$

$$s_{ij,t}^k = \rho_{ij,t}^k \left| P_{tie,ij,t}^k - P_{tie,ij,t}^{k-1} \right| \quad (4.5b)$$

Then, the target variable $P_{target,ij,t}^k$ is updated:

$$P_{target,ij,t}^k = \frac{P_{tie,ij,t}^k - P_{tie,ji,t}^k}{2} \quad (4.6)$$

The Lagrange multipliers are updated using the current tie-line schedule and the updated target:

$$\lambda_{ij,t}^{k+1} = \lambda_{ij,t}^k + \rho_{ij,t}^k (P_{tie,ij,t}^k - P_{target,ij,t}^k) \quad (4.7)$$

To accelerate convergence, an adaptive penalty strategy is employed. The penalty parameter $\rho_{ij,t}^{k+1}$ is dynamically adjusted based on the ratio of the squared primal and dual residuals:

$$\rho_{ij,t}^{k+1} = \begin{cases} \rho_{ij,t}^k \cdot \tau & \text{if } (r_{ij,t}^k)^2 > \mu (s_{ij,t}^k)^2 \\ \rho_{ij,t}^k / \tau & \text{if } (s_{ij,t}^k)^2 > \mu (r_{ij,t}^k)^2 \\ \rho_{ij,t}^k & \text{otherwise} \end{cases} \quad (4.8)$$

where $\mu = 10$ and $\tau = 1.5$ are the adaptive balancing parameters.

Step 4: Convergence Check. The iterative algorithm terminates when both the primal and dual residuals satisfy the convergence tolerance ϵ_{stop} for three consecutive iterations. Specifically, the convergence criteria are defined as:

$$(r_{ij,t}^k)^2 \leq \epsilon_{stop} \quad \text{and} \quad (s_{ij,t}^k)^2 \leq \epsilon_{stop} \quad \text{for 3 consecutive steps} \quad (4.9)$$

where the stopping tolerance is set to $\epsilon_{stop} = 0.001$. If the criteria are met, the algorithm stops and the final optimal scheduling is achieved. Otherwise, the iteration counter is incremented ($k \leftarrow k + 1$) and the process returns to Step 2.

CHAPTER 5. TEMPORAL COORDINATION WITH MODEL PREDICTIVE CONTROL

5.1 Model Predictive Control Time Framework

In the operational context of Multi-Microgrid systems, independent optimization at a single time instance or a fixed time frame is often inadequate to cope with unpredictable fluctuations from load demand ($P_{load,i,t}$, $Q_{load,i,t}$) and renewable energy sources ($P_{PV,i,t}^{peak}$, $P_{WT,i,t}^{peak}$), as well as unexpected events such as main grid disconnection. To address this challenge, Model Predictive Control (MPC) is implemented as the highest-level Temporal Coordination layer.

The MPC operates on a Rolling Horizon principle, continuously solving the optimization problem over a prediction horizon h at each control step, but only applying the control signals of the first step to the actual system. In this proposed architecture, the temporal framework is strictly divided into two primary phases: Day-Ahead Scheduling and Real-Time Emergency MPC.

5.1.1 Day-Ahead Scheduling

During the Day-Ahead Scheduling phase, the optimization problem is formulated and solved for the entire 24-hour cycle of the upcoming day, meaning the prediction horizon is set to $h = 24$ hours. The primary objective is to establish an economically optimal operational plan based on long-term forecasts of renewable power generation ($P_{PV,i,t}^{peak}$, $P_{WT,i,t}^{peak}$) and load demand ($P_{load,i,t}$, $Q_{load,i,t}$).

In this phase, the system determines the baseline charging and discharging schedules for the Battery Energy Storage Systems (BESS) using variables such as $P_{ch,i,t}$, $P_{dis,i,t}$, $u_{ch,i,t}$, and $u_{dis,i,t}$. Furthermore, it finalizes the unit commitment status of backup sources like Diesel Generators (DG) represented by $u_{DG,i,t}$ and $v_{DG,i,t}$, aiming to minimize the overall operational costs while ensuring sufficient spinning reserve capacity ($R_{DG,t}$).

5.1.2 Real-Time Emergency MPC

To address immediate deviations, fault clearing, component isolation, and dynamic responses during emergencies, the Real-Time Emergency MPC is activated. This phase employs a rolling horizon of $h = 5$ hours. At each time step $t \in \mathcal{T}$, the system updates the latest measurements regarding renewable generation, actual load, and the current State of Charge of the BESS ($SOC_{i,t}$). The optimization problem is continuously re-evaluated over the sliding window from t to $t + h - 1$.

During emergency conditions, such as a complete disconnection from the main grid ($P_{grid,t}^{buy} = 0$, $P_{grid,t}^{sell} = 0$), this dynamic mechanism tightly integrates with the optimization algorithms to rapidly re-coordinate the active power traded via tie-lines ($P_{tie,ij,t}$) between interconnected microgrids. By continuously rolling the time horizon, the system effectively updates $SOC_{i,t}$ based on the efficiency η_{BESS} and makes optimal decisions to dynamically handle load shedding. It strategically allocates the shedding between normal ($P_{shed,i,t}^{normal}$) and critical loads ($P_{shed,i,t}^{critical}$) or adjusts BESS and DG operations accordingly, thereby maintaining the stability and resilience of the Multi-Microgrid system under the most severe conditions.

5.2 State Update and Forecasting Mechanism

At each discrete time step t , before the optimization problem is formulated and solved for the upcoming prediction horizon h , the system must inherit the latest physical states from the previous control action and integrate the newly acquired forecasting vectors.

5.2.1 State of Charge Inheritance

In the context of the Multi-Microgrid system, the most critical physical state coupling consecutive time steps is the State of Charge (SOC) of the Battery Energy Storage Systems (BESS). The SOC at the beginning of the new rolling horizon must accurately reflect the state resulting from the control actions implemented in the previous step.

For a prediction horizon starting at the current time step t , the initial SOC parameter, denoted as SOC^{ini} for the current optimization window, is directly updated by the realized SOC from the end of the previous time step $t - 1$:

$$SOC^{ini} = SOC_{i,t-1} \quad \forall i \in \mathcal{N}^{BESS} \quad (5.1)$$

where $SOC_{i,t-1}$ represents the State of Charge of the BESS at node i evaluated at the prior interval. This value serves as the initial boundary condition for the BESS constraints over the new prediction horizon.

Consequently, the SOC transition within the predictive window for any prospective step $\tau \in [t, t + h - 1]$ is formulated based on the charging ($P_{ch,i,\tau}$) and discharging ($P_{dis,i,\tau}$) power variables:

$$SOC_{i,\tau} = SOC_{i,\tau-1} + \frac{\eta_{BESS} P_{ch,i,\tau}}{E_{BESS}} \Delta t - \frac{P_{dis,i,\tau}}{\eta_{BESS} E_{BESS}} \Delta t \quad \forall i \in \mathcal{N}^{BESS} \quad (5.2)$$

where η_{BESS} denotes the charging/discharging efficiency, and E_{BESS} is the total energy capacity of the BESS.

5.2.2 Forecasting Data Update Mechanism

Beyond the state inheritance, the MPC architecture requires continuous refreshing of the anticipated uncontrollable variables, specifically the active and reactive load demand, alongside the renewable energy generation capabilities.

At each control iteration starting at time t , the energy management system retrieves the most recent forecast profiles. These predictions are structured as vectors spanning the entire prediction horizon h . The forecasting vectors updated at step t are formally defined as follows:

- **Load Demand Forecasts:** The active and reactive power demand vectors for each node i are updated as:

$$\mathbf{P}_{load,i}^{(t)} = [P_{load,i,t}, P_{load,i,t+1}, \dots, P_{load,i,t+h-1}]^T \quad (5.3)$$

$$\mathbf{Q}_{load,i}^{(t)} = [Q_{load,i,t}, Q_{load,i,t+1}, \dots, Q_{load,i,t+h-1}]^T \quad (5.4)$$

- **Renewable Energy Forecasts:** Similarly, the maximum available active power from Photovoltaic (PV) systems and Wind Turbines (WT) is integrated into the

control model via vectors:

$$\mathbf{P}_{PV,i}^{(t)} = \left[P_{PV,i,t}^{peak}, P_{PV,i,t+1}^{peak}, \dots, P_{PV,i,t+h-1}^{peak} \right]^T \quad (5.5)$$

$$\mathbf{P}_{WT,i}^{(t)} = \left[P_{WT,i,t}^{peak}, P_{WT,i,t+1}^{peak}, \dots, P_{WT,i,t+h-1}^{peak} \right]^T \quad (5.6)$$

By systematically aligning the inherited SOC^{ini} and the dynamically refreshed forecast vectors $\mathbf{P}_{load,i}^{(t)}$, $\mathbf{Q}_{load,i}^{(t)}$, $\mathbf{P}_{PV,i}^{(t)}$, and $\mathbf{P}_{WT,i}^{(t)}$, the MPC optimization engine is effectively recalibrated at every iteration. This mechanism ensures that the formulated scheduling and trading decisions are robust against uncertainties and tightly bound to the true physical constraints of the distribution network.

5.3 The 3-Mode Spatiotemporal Coordination Framework

The continuous and stable operation of the Model Predictive Control (MPC) strategy inherently relies on a dynamic rolling mechanism that functions as a state machine. To efficiently manage computational resources while ensuring system resilience, the spatiotemporal coordination framework is explicitly structured into three distinct operating modes: Mode 1 (Normal Operation), Mode 2 (Emergency Mode), and Mode 3 (Recovery Mode).

5.3.1 Mode 1 and Mode 3: Normal Operation and Recovery

Under normal operating conditions (Mode 1), where no faults are detected within the distribution network, the real-time dispatch simply tracks the established Day-Ahead schedule. In this state, the actual operational variables strictly follow the day-ahead baseline, meaning the system state $\mathbf{x}_t$ and control actions $\mathbf{u}_t$ satisfy $\mathbf{u}_t = \mathbf{u}_t^{DA}$. This baseline schedule is derived from the 24-hour ATC SOCP optimization problem described in Chapter 4 (Eq. 4.4). Thus, the system operates securely without the need for iterative rolling horizon recalculations, thereby significantly conserving computational resources.

However, when a fault is cleared following an emergency event, the system transitions into Mode 3 (Recovery Mode). Rather than immediately resuming the potentially outdated Day-Ahead schedule, the energy management system recalculates a new optimal operational plan from the current time t_{faults}^{end} to the end of the 24-hour cycle, yielding a shrinking prediction horizon $\mathcal{H}_{rec} = \{t_{faults}^{end}, t_{faults}^{end} + 1, \dots, 24\}$. The updated forecast vectors (Eqs. 5.3–5.6) are integrated into the optimization engine, and the modified ATC Global Objective is solved to establish a new operational baseline:

$$\min \sum_{\tau=t}^{24} \sum_{n \in \mathcal{N}_{MG}} F_{MG,n}(\tau) \quad (5.7)$$

Once this new updated baseline schedule $\mathbf{u}_{\tau}^{DA*}$ is successfully deployed, the system securely reverts to Mode 1.

5.3.2 Mode 2: Emergency Integration of Spatial and Temporal Control

In the face of extreme scenarios such as $N - 2$ contingencies and main grid connection faults (resulting in complete islanding where $P_{grid,t}^{buy} = 0$ and $P_{grid,t}^{sell} = 0$), relying solely on independent spatial or temporal coordination is mathematically and

physically insufficient. The system enters Mode 2 (Emergency Mode), employing a bottom-up integration strategy that intrinsically couples the ATC spatial algorithm with the MPC rolling horizon framework. This synergistic approach allows the Multi-Microgrid system to maintain rigorous operational resilience under critical conditions.

The integration mechanism is formally structured such that the MPC layer acts as the primary temporal coordinator, dictating the rolling horizon scheduling. Within each time step t of the MPC's sliding predictive window $\mathcal{H}_{MPC} = \{t, t+1, \dots, t+h-1\}$, the ATC spatial algorithm, formulated as a convex Second-Order Cone Programming (SOCP) problem, is iteratively invoked.

When a contingency occurs, the dynamic topology and physical network constraints are instantly updated. For each prospective control interval $\tau \in \mathcal{H}_{MPC}$, the MPC provides the temporal coupling via the updated State of Charge inheritance $SOC_{i,\tau}$ (Eq. 5.2), alongside the most recent forecasting vectors for load demand and renewable generation ($\mathbf{P}_{load,i}^{(t)}, \mathbf{Q}_{load,i}^{(t)}, \mathbf{P}_{PV,i}^{(t)}, \mathbf{P}_{WT,i}^{(t)}$). The ATC algorithm then iteratively optimizes the spatial power distribution and active power traded via inter-microgrid tie-lines ($P_{tie,ij,\tau}$) by minimizing the ATC Global Objective (Eq. 4.4). The distributed coordination mathematically relies on updating the Augmented Lagrangian multiplier $\lambda_{ij,\tau}$ and the penalty parameter $\rho_{ij,\tau}$ to reach an optimal consensus on the target traded active power $P_{target,ij,\tau}$ designated by the DSO. At each iteration k , the multiplier update step is executed as:

$$\lambda_{ij,\tau}^{(k+1)} = \lambda_{ij,\tau}^{(k)} + \rho_{ij,\tau} \left(P_{tie,ij,\tau}^{(k)} - P_{target,ij,\tau} \right) \quad \forall \tau \in \mathcal{H}_{MPC} \quad (5.8)$$

By explicitly embedding the ATC SOCP solver within the outer MPC loop, the control system mathematically guarantees that both the inter-temporal energy constraints and the spatial power flow limits—including the tightened apparent power flow bounds and nodal voltage magnitude constraints ($V_{min}^2 \leq V_{i,\tau}^2 \leq V_{max}^2$)—are strictly satisfied at every temporal stage, even when the network topology is severely compromised due to multiple cascading line outages.

5.3.3 Activation of Penalty Functions and Load Shedding Mechanism

During emergency mode operations, the available generation capacity from local Diesel Generators ($P_{DG,i,t}$), BESS discharging ($P_{dis,i,t}$), and renewable resources may fail to cover the aggregated demand. Consequently, the load shedding mechanism is explicitly activated as a last resort to preserve overall system stability.

The optimization objective incorporates severe penalty functions to strategically manage energy deficits and state deviations. The total active power load shedding at each node $P_{shed,i,t}$ is decoupled into normal load shedding ($P_{shed,i,t}^{normal}$) and critical load shedding ($P_{shed,i,t}^{critical}$). The objective function dynamically penalizes these curtailments using distinct Value of Lost Load (VOLL) coefficients: a penalty cost for normal load shedding C_{shed}^{normal} and a significantly higher penalty cost for critical load shedding $C_{shed}^{critical}$. The load shedding cost function is formulated as:

$$J_{shed,\tau} = \sum_{i \in \mathcal{N}} \left(C_{shed}^{normal} P_{shed,i,\tau}^{normal} + C_{shed}^{critical} P_{shed,i,\tau}^{critical} \right) \quad (5.9)$$

This strict prioritization ensures that critical infrastructure remains energized while flexible loads are curtailed first.

Furthermore, to maintain energy reserves for extended islanded operation, a terminal State of Charge penalty is activated. By introducing a slack variable $S_{SOC,i}$ and a penalty coefficient β_{SOC} , the MPC penalizes the deviation of the BESS SOC at the end of the predictive horizon from a secure reference level. The SOC penalty function is given by:

$$J_{SOC} = \sum_{i \in \mathcal{N}^{BESS}} \beta_{SOC} S_{SOC,i}^2 \quad (5.10)$$

This temporal penalty works in tandem with the spatial penalty coefficient for distribution network power losses ω_{loss} , preventing aggressive discharging that would leave the BESS depleted ($SOC_{i,t}$ approaching SOC^{min}) for subsequent intervals.

Through this cohesive integration, the ATC SOCP algorithm resolves spatial power flow conflicts while the MPC framework optimizes the inter-temporal energy buffering and load shedding deployment, enabling the Multi-Microgrid system to navigate $N - 2$ contingencies and main grid disconnections seamlessly.

Algorithm 1 Bottom-Up Spatiotemporal Coordination as a 3-Mode State Machine (Part 1)

Require: Real-time network measurements, BESS characteristics, Forecast models.

- 1: **DAY-AHEAD SCHEDULING PHASE**
 - 2: Fetch long-term 24-hour forecasts $\mathbf{P}_{load}^{DA}, \mathbf{P}_{PV}^{DA}, \mathbf{P}_{WT}^{DA}$
 - 3: Solve ATC Global Objective for the entire 24-hour horizon ($h = 24$)
 - 4: Establish Day-Ahead baseline schedule $\mathbf{u}^{DA}$ for $t \in [1, 24]$
 - 5: **REAL-TIME EMERGENCY MPC PHASE**
 - 6: **for** each time step $t = 1$ to 24 **do**
 - 7: **STEP 1: STATE UPDATE**
 - 8: Read current BESS status and inherit SOC^{ini} according to Eq. (5.1)
 - 9: **STEP 2: MODE BRANCHING & EVENT DETECTION**
 - 10: **if** No fault **then**
 - 11: **Mode 1 (Normal Operation):**
 - 12: Extract data and decisions from Day-Ahead baseline: $\mathbf{u}_t = \mathbf{u}_t^{DA}$
 - 13: Apply scheduled control signals for step t
 - 14: **else if** Fault detected at current hour t **then**
 - 15: **Mode 2 (Emergency Mode):**
 - 16: Horizon $h \leftarrow 5$ $\triangleright$ Note: Forecast vectors seamlessly extend beyond $t = 24$ for late-day windows
 - 17: Fetch rolling forecast vectors $\mathbf{P}_{load,i}^{(t)}, \mathbf{Q}_{load,i}^{(t)}$ (Eq. 5.3–5.4)
 - 18: Fetch renewable potential $\mathbf{P}_{PV,i}^{(t)}, \mathbf{P}_{WT,i}^{(t)}$ (Eq. 5.5–5.6)
 - 19: **Activate Emergency Mode Constraints** ($\Gamma_E(\tau) = 1$):
 - 20: • Isolate mathematically identified faulty components. (3.10)
 - 21: • Enforce tightened nodal voltage bounds. (3.11e)
 - 22: • Activate load shedding allowance. (3.8a)–(3.8c)
 - 23: **Update Objective Function:** Switch to $\mathcal{F}_{emergency}^{OPF}$ (3.15) and add SOC penalty J_{SOC} (5.10)
-

Algorithm 1 Bottom-Up Spatiotemporal Coordination as a 3-Mode State Machine (Part 2)

Algorithm 1 (Continued) – MODE 2 (Emergency Mode) Continued

24: **STEP 3: ROLLING HORIZON OPTIMIZATION LOOP**

25: **Call** Spatial ATC Algorithm to optimize spatial power flow

26: Compute SOC transitions simultaneously using Eq. (5.2)

27: Iteratively solve ATC Global Objective (4.4) for period $\tau \in [t, t + h - 1]$.

28: Enforce tie-line target exchange agreement $\triangleright$ Solver intrinsically handles power deficits via shedding penalties

29: **STEP 4: CONTROL IMPLEMENTATION**

30: Extract optimized decision variables for the entire horizon $\tau \in [t, t + h - 1]$

31: **Apply only** the control signals for the first step $\tau = t$ to physical assets (BESS, DG, Loads)

32: **else if** Fault cleared at current step t **then**

33: **Mode 3 (Recovery Mode):**

34: Horizon $h \leftarrow 25 - t$

35: Fetch forecast vectors $\mathbf{P}_{load,i}^{(t)}, \mathbf{Q}_{load,i}^{(t)}$ (Eq. 5.3–5.4)

36: Fetch renewable potential $\mathbf{P}_{PV,i}^{(t)}, \mathbf{P}_{WT,i}^{(t)}$ (Eq. 5.5–5.6)

37: Iteratively solve ATC Global Objective (4.4) for period $\tau \in [t, 24]$.

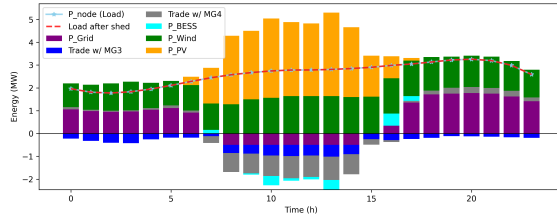
38: Update Day-Ahead baseline schedule $\mathbf{u}^{DA}$ with newly calculated decisions

39: Extract and apply control signals for step t from the new baseline

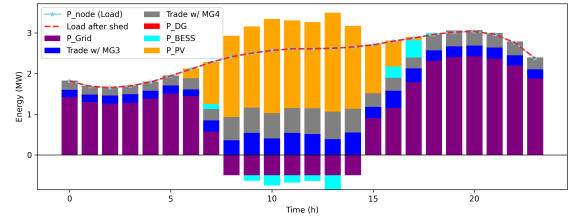
40: **end if**

41: **end for**

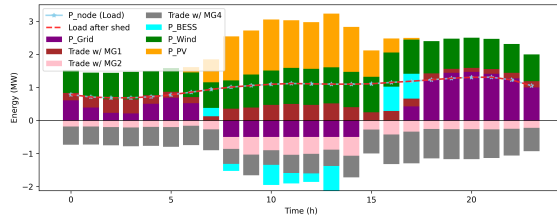
CHAPTER 6. RESULTS AND DISCUSSION



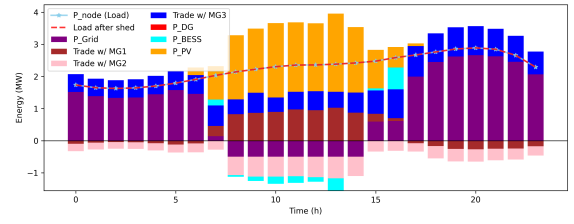
(a) Active Power MG1



(b) Active Power MG2



(c) Active Power MG3



(d) Active Power MG4

Hình 6.1 Test 2x2 Layout: Default Active Power for 4 Microgrids.

CHAPTER 7. CONCLUSION

LIST OF SCIENTIFIC PUBLICATIONS

- **Trinh Xuan Thanh**, [Co-author(s)], Nguyen Duc Tuyen, “[*Paper Title*]”, [Journal/Conference Name], 2025 (Under Review / Accepted / Published).

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